

A Critical Review of Operational Strategies for the Management of Harmful Algal Blooms (HABs) in Inland Reservoirs

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Abstract

Occurrences of freshwater harmful algal blooms (HABs) are increasing on a global scale, largely in part due to increased nutrient input and changing climate patterns. While reservoir management strategies that can influence phytoplankton are known, there is no published guideline or protocol for the management of harmful algal blooms. There is a need to establish what factors are the predominant drivers of blooms, and how common reservoir management strategies specifically influence each factor. The following literature review seeks to establish the benefits and drawbacks of operational management strategies that currently exist. The main focus is altering hydrodynamic conditions (hypolimnetic withdrawals, surface flushing, pulsed inflow, artificial mixing, and varying water level), in order to induce environmental changes within the reservoir itself. Since excess nutrients are one of the biggest contributors to worsening bloom conditions, internal nutrient dynamics and reduction are also discussed. Additionally, we review the predominant seasonal factors (stratification, light, temperature, and wind) that affect likelihood of bloom occurrence and duration. The ultimate objective of this review is to increase understanding of the relationships between HAB drivers and reservoir operations in order to inform the development of data, modeling, and management strategies for the prevention and mitigation of blooms.

Keywords

Harmful algal blooms; Reservoir management; Hydrodynamic control; Nutrient dynamics; Seasonal limnology

Introduction

As climate shifts with time and anthropogenic input, temperature and precipitation patterns are expected to change as well. These alterations impact, among other factors, water level

fluctuations, nutrient dynamics, and community composition of algal species within reservoirs. As population increases, a higher demand of reservoir water for drinking and agriculture will be experienced. To successfully balance water quantity and quality demands, reservoir operations must anticipate and adapt to these and other factors outside local control.

One of the primary goals of this paper is to establish the strengths and weaknesses of common methods of bloom control. Water resource managers can then use this information to determine which strategy is best suited for improving water quality within their system. Since variation of hydrodynamic conditions within a reservoir itself is often the most feasible method of control, this is discussed first. Section 1 includes hypolimnetic withdrawals, surface withdrawals, pulsed inflow, artificial mixing, and water level fluctuations. These hydrodynamic methods have significant impact on other limnological features, such as nutrient dynamics, light availability, stratification, and temperature. Section 2 covers relationships between phosphorus and nitrogen, nutrient co-limitation with light, as well as several possible methods for nutrient reduction. It is important to note that many of the hydrodynamic methods discussed in Section 1 are applied to decrease blooms during peak periods and should be coupled with an additional method of nutrient reduction in order to produce longer-term results. Section 3 discusses several factors that change seasonally: stratification, light, temperature, water level, and wind. It is also important to understand the dynamic nature of blooms, therefore the final section highlights how seasonal variation, as well as how climate change is expected to impact critical factors and influence reservoir water quality. Given projected worsening bloom conditions, proficient and proactive management of reservoirs will be required to keep blooms under control.

1. Hydrodynamic condition control

Changing hydrodynamic conditions within a reservoir is often a low cost, feasible method of controlling HABs. When determining what type of adaptive management strategy to choose, multiple factors must be taken into consideration such as physical morphology of the system, nutrient dynamics, timing and distribution of blooms, extent of stratification, water level, required outflow conditions, as well as other indicators of watershed and reservoir water quality.

1.1 Hypolimnetic withdrawals

Hypolimnetic withdrawals are a water quality restoration strategy in which bottom water is selectively discharged out of the system. This method works mainly through reduction of bottom water nutrient concentrations, which over time can lead to reduction of surface nutrients. This leads to nutrient limitation of phytoplankton and a decrease in blooms. This method is most suited for deep, strongly stratified reservoirs. Withdrawals are most effective in

the late summer or early fall, after bottom nutrients have accumulated. When using this method, it is important not to break up stratification too soon, or high levels of nutrients will become available to the surface, worsening blooms. If carried out correctly, this is a low cost, simple method of nutrient control that can effectively reduce frequency and duration of HABs.

During peak stratification periods, oxygen from the surface is not mixed down and anoxic conditions persist in bottom layers. Under these conditions, systems with high legacy phosphorus often release phosphorus from the sediment into the water. The use of hypolimnetic withdrawals removes this excess phosphorus, preventing it from being mixed to the surface. Long term, continuous export of hypolimnetic, nutrient rich waters can lead to an overall decrease in surface total phosphorus concentrations, and reduction of harmful blooms (Nurnberg, 2020; Nurnberg, 2007). Summer average phosphorus decreases were found to be strongly correlated with annual water volumes and phosphorus masses withdrawn per lake area, demonstrating the importance of timing of the treatment and hydrologic conditions (Nurnberg, 2007). Timing and amount of withdrawal is important to maintaining healthy water quality within the system. For best success, timing of withdrawals should coincide with higher summer internal phosphorus loads and occur before the onset of destratification in the fall (Nurnberg, 2020; Bormans et al., 2015; Dunalska et al., 2007).

The use of hypolimnetic withdrawals also reduces total nitrogen and total iron concentrations significantly (Dunalska et al., 2007). After several years, this method can decrease legacy nutrients in sediments and improve overall ecosystem health. Often with withdrawals and destabilization of the water column, community composition shifts to less noxious species (Nurnberg, 2007). Typically, this results in a change in dominance from toxic cyanobacteria to less harmful diatoms (Bormans et al., 2015).

In some cases, water quality remained the same or worsened due to hypolimnetic withdrawal, typically due to significant temperature increase in the hypolimnetic layer. Withdrawals that are too vigorous or incorrectly timed can lead to destratification during periods of high light and warm water temperature, creating conditions more conducive to cyanobacterial blooms. Under these conditions, nutrient rich bottom water becomes mixed with surface water, leading to more nutrient availability to surface cyanobacteria and an adverse effect on water quality (Bormans et al., 2015; Nurnberg, 2007). The timing and extent of withdrawal are critical and must direct the operation of the treatment. In particular, timing and rates of withdrawal must be managed to prevent early onset destratification and subsequent radical bottom layer temperature increases (Silvonen et al., 2021; Nurnberg 2007). Hypolimnetic withdrawals are also unsuccessful if total phosphorus concentrations are so high they remain above the threshold for phosphorus limitation. Even when successfully carried out, hypolimnetic withdrawals produce cold and phosphorus rich outflow. This can have several adverse effects on downstream waters, such as eutrophication, cold-water pollution, oxygen depletion, and odor development (Weber et al., 2017; Nurnberg, 2007). Effect of water quality on downstream

species, as well as potential treatment of outflow, should be considered when using this method.

Hypolimnetic withdrawals can be implemented through a few methods. A simple, low cost method is to install a pipe at the bottom of the reservoir, with the ability to control the rate of flow with seasons. These can be constructed from fiberglass and polyester resin (Dunalska et al., 2007). Another method is the use of a top temperature control curtain. This works by using a curtain to contain the upper layers of water, facilitating high water flow below the epilimnion (He et al., 2017). This method is discussed in more detail in Section 3. Hypolimnetic withdrawals are only a feasible option for systems with a deep enough water level and large enough water load to support the volume of withdrawal required for effective treatment during peak periods (Nurnberg, 2020). Systems that benefit most from this management strategy are strongly stratified and poorly mixed, with high accumulation of nutrients in bottom water layers. This method, if used correctly, is a low cost, feasible solution to control both nutrient levels and HABs within a reservoir.

1.2 Horizontal flushing

Horizontal flushing is a water quality restoration strategy in which surface water is selectively discharged out of the system. Surface withdrawals reduce HABs primarily by directly removing bloom biomass and exporting it out of the system (Rigosi and Rueda, 2012). Additionally, this method decreases surface temperature and nutrient concentration. This method is typically most effective during summer months, when stratification is strongest and large blooms are likely to occur. Since high flow rate cannot be maintained indefinitely, it is important to determine when and for how long high velocity flow is needed in order to best combat HABs. Typically, short, intermediate disturbance events are ideal. This is a short term strategy suited for decreasing blooms during peak periods, and should be coupled with an additional method of nutrient reduction in order to produce longer-term results.

Increase in hydraulic flushing has been shown to inhibit or break up cyanobacterial blooms (Roelke and Pierce, 2011; Roelke et al., 2010; Moustaka-Gouni et al., 2006). Surface nutrient concentrations, primarily phosphorus, can be effectively controlled with shallow water withdrawals, helping reduce HABs. Additionally, by removing organic matter produced in the upper water layers, the amount of nutrients reaching the sediments decreases, thus reducing the overall nutrients available for later resuspension (Carr et al., 2020). This method is typically ideal for systems that have seasonal blooms during distinct periods, and where natural temperature needs to be restored downstream. Withdrawing water from the surface can be advantageous for reducing epilimnion temperature, thus reducing extent of blooms, particularly during flood conditions. However, while changing environmental factors within the reservoir is important, horizontal flushing works mainly through dilution and direct removal from the system.

Nevertheless, horizontal flushing does contain several drawbacks. Removal of oxygen rich surface water can worsen anoxic conditions, since water left behind in the reservoir will be depleted in oxygen (Weber et al., 2017). This method also comes with the tradeoff of increased phosphorus and nitrogen transport downstream (Carr et al., 2020). Harmful toxins can be transported downstream and damage downstream species (Herman et al., 2017). Treatment of outflow should be considered when using this method. Additionally, determining flushing thresholds beyond which harmful taxa are unable to form blooms is complex and must be taken into account for each individual system (Roelke and Pierce, 2011).

Horizontal flushing is typically carried out through the use of an outlet near the top of the reservoir, with flow rate controlled by a pump. Another method is the use of a bottom temperature control curtain (discussed in more detail in Section 3), ultimately causing the reservoir to behave similar to a river channel (He et al., 2017). Duration and frequency of horizontal flushing is essential. If flow is maintained for too long, it can lead to the development of a low diversity phytoplankton population adapted to these specific conditions (Rigosi and Ruenda, 2012). A requirement for continuous flushing is high inflow limiting its use in systems with strict water budgets. Short disturbances of intermediate frequency are more ideal to induce environmental changes, promoting higher diversity and reducing blooms dominated by a specific species (Rigosi and Ruenda, 2012; Padisak et al., 1999). Intermediate pulses can be defined as every 20-30 days, with the intensity of 1%-2% of reservoir volume per day (Padisak et al., 1999).

1.3 Pulsed inflow

Pulsed inflows utilize infrastructure such as weirs to release upstream water specifically to increase flow rate in downstream areas of the reservoir and combat HABs. This is a short-term method that decreases HABs through disruption of stratification, light limitation, and direct dilution of cyanobacterial biomass. This method is most suited for riverine-like systems, where blooms typically occur seasonally in distinct portions of the reservoir. However, breaking up stratification during peak bloom periods is not an effective long term management strategy. In order to better reduce HABs, pulsed inflows should be coupled with a nutrient reduction management strategy.

Pulsed inflows primarily reduce HABs through the dilution and relocation of cells, light limitation, and suppression of persistent thermal stratification (Xu et al., 2021). Pulsed inflows are important to consider when dealing with narrow, riverine-like reservoirs that have the ability to release water from upstream storage to flush and dilute blooms downstream (Li et al., 2021; Xu et al., 2021). High masses of inflow have been found to effectively break up stratification, alter temperature profiles, and reduce phytoplankton biomass. For some systems, breaking up stratification during peak periods and inducing mixing is key to suppressing HABs. In stratified systems, nutrient limitation at the surface allows slow growing

cyanobacteria to dominate, and high light environments typically coincide with higher cell counts of cyanobacteria (Mitrovic et al., 2011). Pulsed inflows increase turbidity in the water column, entraining buoyant cyanobacteria where less light is available, decreasing their growth, and resulting in less cyanobacterial cell counts (Lian et al., 2014; Mitrovic et al., 2011). Mixing also increases the growth rate of negatively buoyant algae through reduced sediment loss, allowing them to compete at the surface (Visser et al., 2016). Light becomes the limiting factor, and algae better adapted to fluctuating light levels begin to co-dominate

Pulsed inflows require a high discharge, limiting availability for other water resource users, such as those needed for power generation. Water supply demand must be taken into account when using this method (Li et al., 2021; Mitrovic et al., 2011). Water depth must also be considered. Shallower water depth favors surface floatation of cyanobacteria and bloom formation (Zhang et al., 2021). Additionally, if releases significantly decrease upstream water level, it can lead to increased algal cell density in these areas, especially in shallower systems (Xia et al., 2020). Flow velocity and timing of release must be determined for each individual system based on bathymetry, cross sectional area, meteorological conditions, and nutrient dynamics within the system. A critical velocity of 0.05 m/s has shown success suppressing persistent stratification and *Anabaena* blooms (Mitrovic et al., 2003) and holds potential for application to rivers in similar climate zones. Flow rate should also be altered based on seasonality and past bloom extent. The use of continuous, lower velocity flow can often be used to prevent blooms. However, once a bloom has occurred, a larger velocity must be induced to disrupt it. Surface flow rate in areas with low velocity and high temperatures should be increased during periods when temperature stratification is strongest (Xu et al., 2021; Chuo et al., 2019). Meteorological trends can be used to predict when the effects of pulsed inflows on HABs can be maximized (Xin et al., 2020). Studies have shown phytoplankton biomass has been reduced through increase of flow rate when forecasted temperature is projected to be continuously higher (three degrees Celsius or greater) than the historical 5 day mean value of the corresponding period (Xin et al., 2020). If timed correctly, increasing flow rate during vulnerable periods is an effective, low cost method to reduce HABs on a short term basis.

1.4 Artificial mixing

Artificial mixing breaks up stratification and can be successful at reducing blooms in which light is the limiting factor to growth. There are a few types of artificial mixers available (Visser et al., 2016). Aeration systems are designed to subject air to high pressure, induce mixing, and break up stratification. Selection of the inlet is crucial, with a deeper inlet typically producing more effective mixing. Mechanical destratification devices work through using a pump to decrease temperature difference across the thermocline, either by pumping cold water upwards or warm water down. Two different types of pumps are commonly used, an axial pump with a large propeller to generate low velocity jets, or a direct drive mixer with a small propeller to generate high velocity jets. Mechanical mixing is used less frequently because it is not as effective as

aeration. Solar-powered mixers improve surface flow by pumping water to the surface using an intake hose and then transporting it radially. These mixers can only cover a small area and typically are not adequate to mix entire lakes. Artificial mixing is typically not successful in shallower systems (<5 m) or those with high levels of legacy nutrient sediment release. In fact, mixing can often make blooms worse in these cases.

Artificial mixing has been successful in some systems at reducing overall phytoplankton biomass, as well as shifting dominance from cyanobacteria to flagellates, green algae, and diatoms (Visser et al., 2016; Jungo et al., 2001). Artificial mixing can also increase bottom oxygen levels, reducing hypoxia and leading to less nutrient release from sediments (Ma et al., 2015; Cong et al., 2011). Mixing pushes positively buoyant cyanobacteria into deeper water, decreasing their light availability and growth (Ma et al., 2015; Lian et al., 2014). Additionally, negatively buoyant species are brought to the surface, increasing competition. This can lead to a shift in community composition in well mixed reservoirs (Visser et al., 2016; Heo and Kim, 2004; Hawkins and Griffiths, 1993). However, while effective at reducing cyanobacterial blooms specifically, artificial mixing may not decrease total algal concentration (Heo and Kim, 2004; Hawkins and Griffiths, 1993).

This method is typically most effective at reducing blooms that are light limited. If the phytoplankton are nutrient limited, mixing can actually increase biomass due to increasing nutrient availability to the surface (Jungo et al., 2001). In order for this method to be successful, mixing rate must be high, mixing must be deep enough to limit light availability, and mixers must be distributed to induce turbulent flow in a large portion of the system. As with pulsed inflows, there are depth limitations to using this method. Artificial mixing is typically more effective on deeper reservoirs with uniform structure than on systems with shallower water or more complex bathymetry (Visser et al., 2016). This method does not typically work for shallower systems (less than 15 meters), since water is not deep enough to cause light limitation. The number of successful studies on artificial mixing is limited, and there is high variation in the devices used for mixing and how they are positioned in water systems. Success of mixing varies extensively with environmental factors and must be tested for each individual system to determine effectiveness (Jungo et al., 2001; Visser et al., 2016).

1.5 Summary of hydrodynamic methods

When looking at control of HABs through changing hydrodynamic conditions, there are several management strategies to consider. Table 1 below breaks down several aspects of each method.

Hypolimnetic withdrawals are ideal for maintaining thermal stability, reducing legacy nutrients, and providing cooler outflow. Horizontal flushing is ideal for maintaining thermal stability, reducing surface HAB biomass, maintaining colder hypolimnion temperature, and reducing temperature difference between inflow and outflow. Pulsed inflows are ideal for riverine

systems, where water from upstream can be used to dilute blooms downstream during peak periods. The three above methods also impact water level within the reservoir, which can further alter light availability and nutrient dynamics. Artificial mixing has been successful in some deeper systems at reducing overall phytoplankton biomass, as well as shifting dominance from cyanobacteria to less harmful species. However, artificial mixing shows high variability between systems. Each of these methods should be used with caution, especially in regard to proper timing and impact on overall reservoir health.

Table 1. Breakdown of hydrodynamic adaptive management strategies

	Hypolimnetic Withdrawals	Horizontal Flushing	Pulsed Inflow	Mixing
Management strategy	Removal of hypolimnetic layer	Removal of surface layer	Upstream water used to break up downstream blooms	Use of artificial mixing to entrain cyanobacteria
Works by	Reducing nutrients	Direct biomass removal	Dilution of cells and altering environmental conditions	Light limitation
Stratification	Retains stability	Retains stability	Breaks up	Breaks up
Nutrients	Decreases legacy nutrients released from sediments	Decreases surface nutrients	Alters nutrient distribution	Alters nutrient distribution
Temperature	Temperature increase in hypolimnetic layer	Hypolimnetic temperature retained, surface temperature reduced	Decrease surface temperature, more uniform profile	More uniform temperature profile
Outflow	Cold, oxygen depleted, nutrient rich	Warm, oxygen rich, eutrophic	N/A	N/A
Water Level	Decreases	Decreases	Decreases in some areas, increases in others	N/A
Figure	Supplemental 1	Supplemental 1	Supplemental 2	Supplemental 3
Real Life Applications	Carr et al., 2020; Nurnberg, 2020; Weber et al., 2017; Bormans et al., 2015; Dunalska et al., 2007; Nurnberg, 2017	Carr et al., 2020; Weber et al., 2017; Rigosi and Ruenda, 2012	Shin et al., 2022; Xu et al., 2020; Mitrovic et al., 2011; Buyukates and Roelke, 2005; Mitrovic et al., 2003	Ma et al., 2015; Hudnell et al., 2010; Heo and Kim, 2004; Jungo et al., 2001; Hawkins and Griffiths, 1993

2. Nutrients

Excess nutrients is one of the biggest contributors to worsening bloom conditions. While decreasing amount of nutrients being put into freshwater systems would be ideal, this is often not feasible and/ or out of the control of reservoir managers. Instead, methods of reduction of nutrients already present within reservoirs are needed to control HABs. While control of phosphorus has been the forefront of freshwater systems in the past, more recently, the impact of nitrogen dynamics has been considered more significantly. In addition, co-limitation between light and nutrients must be considered to determine when management strategies would be most effective. Management of nutrients is a key factor of controlling HABs, especially as population grows, increasing our demand for agricultural practices and their subsequent nutrient input.

2.1 Phosphorus versus nitrogen

Phosphorus is typically considered the nutrient limiting to freshwater HABs. However, co-limitation can occur at higher phosphorus levels. Which nutrient is limiting, and to what extent, is specific to the type of lake, redox reactions occurring, and type of phytoplankton. Since nitrogen is typically significant only if higher levels of phosphorus are present, reduction of phosphorus is often the more cost effective and feasible nutrient to control. At higher levels of phosphorus, external nitrogen loading becomes more important and should be considered. Nutrient ratios are an effective tool for assessing potential nutrient limitation (Paerl et al., 2016; Scott et al., 2013; Dzialowski et al., 2005). Guidelines to determining nutrient of focus can be found in the figure below.

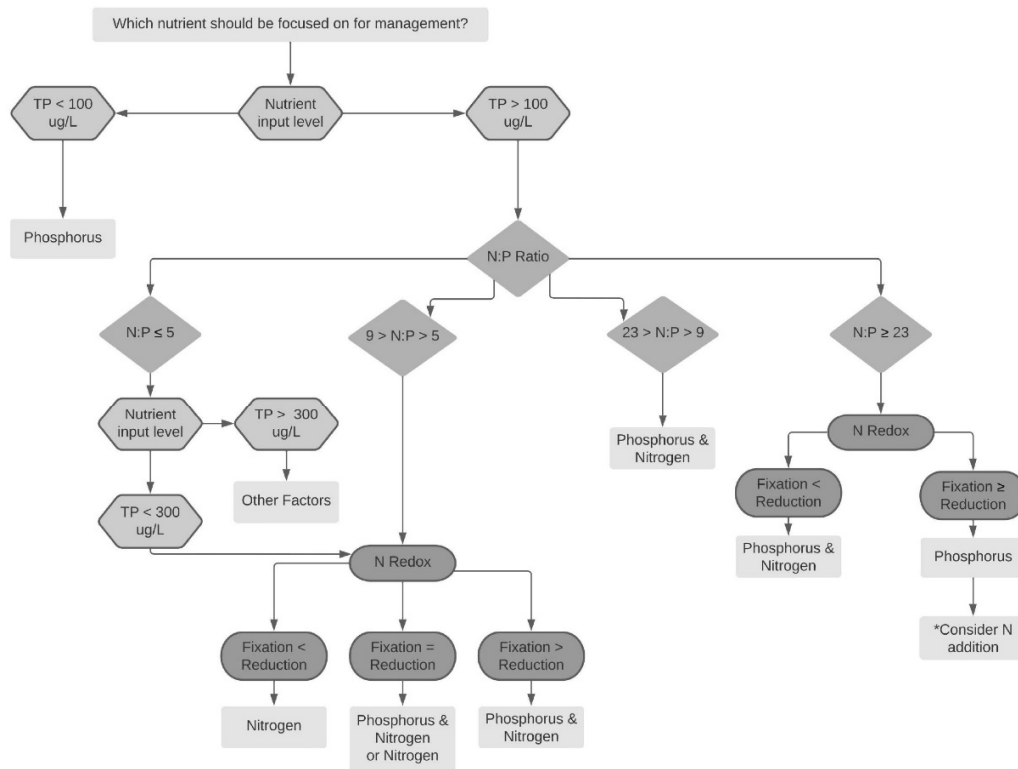


Figure 1. Determination of primary nutrient concern, based on extent of eutrophication, nutrient input, and redox reactions occurring. Constructed from: Filstrup and Downing (2017), Paerl et al. (2016), Yuan and Pollard (2014), Scott et al. (2013), Dzialowski et al. (2005).

For oligotrophic to slightly eutrophic lakes (TP < 100 ug/L), phytoplankton biomass was found to be unrelated to total nitrogen, and instead showed an increasing trend with addition of total phosphorus (Filstrup and Downing, 2017). Decreasing nitrogen in these systems would not be an effective method of control, and it would be more feasible and cost effective to focus on phosphorus reduction only. One way this can be done is through adding iron or alum to sequester phosphorus in sediments, inhibiting phosphorus recycling. This has been found to significantly reduce chlorophyll levels (Schindler et al., 2016).

In hypereutrophic lakes (TP > 100 ug/L), phytoplankton biomass was found to increase with both addition of phosphorus and nitrogen (Filstrup and Downing, 2017). For these systems, it is important to consider the total nitrogen: total phosphorus ratio, and which redox reactions are occurring. Nitrogen should be focused on when nitrogen: phosphorus is less than nine and denitrification exceeds nitrogen fixation (Paerl et al., 2016). If phosphorus is in excess (TP > 300 ug/L) and nitrogen: phosphorus ratio is very low (N:P < 5), factors other than nutrient limitation (such as temperature control) should be considered (Yuan and Pollard, 2014).

In hypereutrophic systems with nitrogen: phosphorus ratios between 9-23, phosphorus and nitrogen co-limit, regardless of dominant redox reaction occurring. Both nutrients should be reduced for these types of systems. Often, in nutrient co-limited lakes, phosphorus reduction is primarily focused on. However, the “P only” control assumes that the nitrogen needed for a system will be supplied by nitrogen fixation. Nitrogen fixation cannot meet the demand for all ecosystems, especially those where denitrification is high and/or legacy phosphorus is high.

Before determining which nutrient reduction strategy to use, it's important to evaluate the algal community composition. In certain systems, the reduction of phosphorus only leads to a shift to toxic species who thrive in high external nitrogen, low external phosphorus environments (Kim et al., 2020). This has been seen in Lake Erie, whose recent surge in *Microcystis* blooms has brought much debate as to whether controlling phosphorus only shifts algal community composition to more toxic strains (Wurtsbaugh et al., 2019). Nitrogen should also be reduced in systems that favor toxic species of non-N fixing cyanobacteria, such as *Microcystis*. *Microcystis* are typically favored in nutrient rich systems with high nitrogen: phosphorus ratios, high water temperature, and high extent of thermal stratification. They especially thrive in areas with low surface phosphorus and high phosphorus release in sediments, due to their ability to migrate through the water column. Here, denitrification typically exceeds nitrification, so external sources of nitrogen become more important. In order to effectively reduce blooms in these systems, both nitrogen and phosphorus must be controlled (Kim et al., 2020; Gobler et al., 2016). The use of hypolimnetic withdrawals and controlled floating wetlands are potential management options to control excess levels of both nutrients.

Strict phosphorus limitation occurs in hypereutrophic lakes when the nitrogen: phosphorus ratio is greater than twenty-three and if denitrification is less than nitrogen fixation (Paerl et al., 2016). Here, phosphorus limitation should be focused on. When reduction of total phosphorus is not possible, addition of total nitrogen can be a short-term solution to switch to more favorable phytoplankton species (Schindler et al., 2016). Larger nitrogen: phosphorus ratios (>50-75:1) allow for chlorophytes to outcompete N-fixing cyanobacteria (Harris et al., 2014). At higher nitrogen: phosphorus ratios, some cyanobacterial blooms are suppressed. This is because non-nitrogen fixing phytoplankton can compete if enough nitrogen is available. Additionally, higher nitrogen: phosphorus ratios leads to larger zooplankton, increasing energy transfer throughout the food web (Harris et al., 2014). This method should be used with extreme caution, and only should be considered if nitrogen fixation exceeds denitrification and non-toxic non-diazotrophs dominate. If nitrogen: phosphorus is high due to external sources of nitrogen, then total nutrients should be reduced to avoid blooms of toxic non-diazotrophs, such as *Microcystis* (Gobler et al., 2016; Kim et al., 2020). Additionally, adding nitrate leads to high redox conditions, preventing mobilization of phosphorus recycled from sediments (Schindler et al., 2016).

2.2 Co-limitation with light

In addition to co-limitation between phosphorus and nitrogen, co-limitation between nutrients and light availability must be considered. If structure of the reservoir permits, decreasing light availability is often a more cost effective and attainable method of controlling blooms than nutrient control (Burrell et al., 2014; Hutchins et al., 2010). Light and nutrient co-limitation can essentially be broken down into three general trends (Figure 2). When both light and nutrients are low, algal growth rate is low and water quality is better. When light is sufficient, nutrient availability will be responsible for growth rate. When nutrients and light are present in excess, they can become limiting. Growth decreases, but water quality is poor due to excess nutrients in the water (Dou et al., 2019). Co-limitation of light and nutrients depends primarily on seasonality and structure of the reservoir.

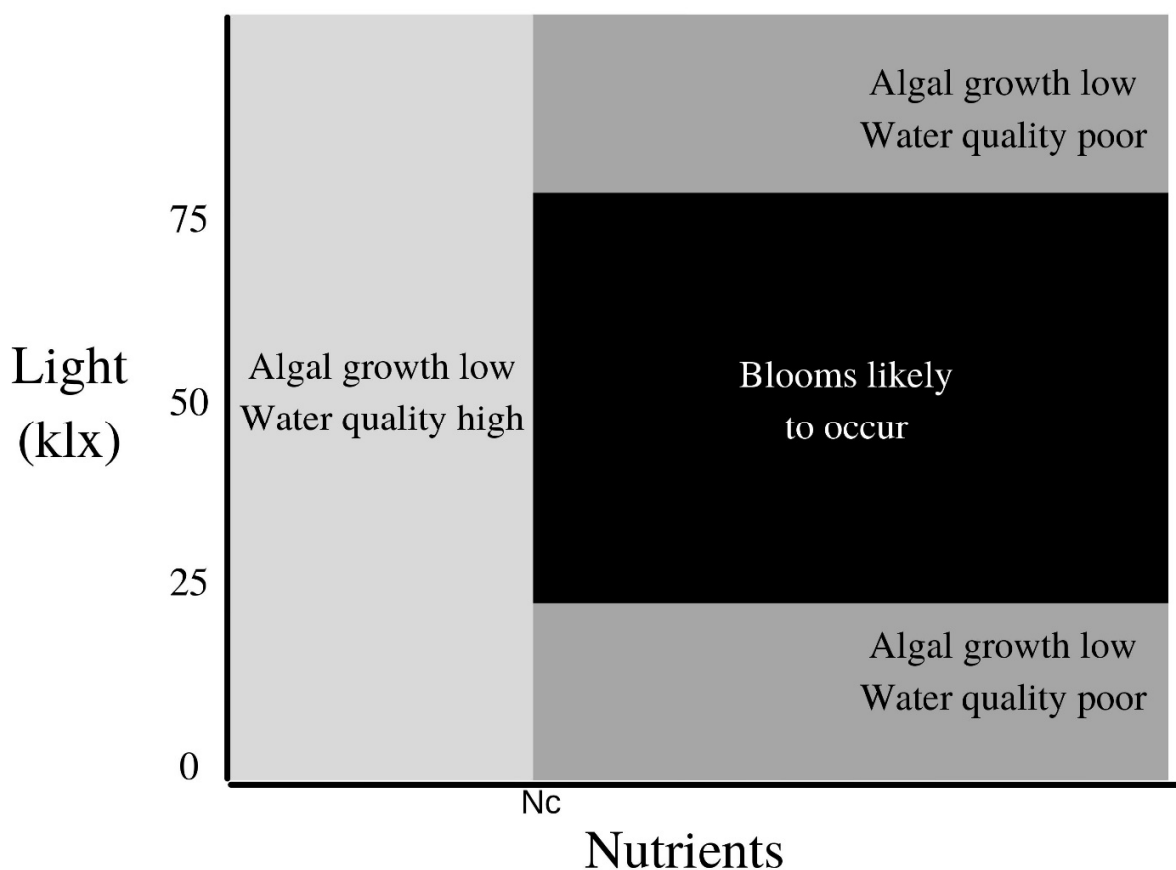


Figure 2. Co-Limitation between light and nutrients. A conceptual figure showing algal growth and water quality based on light and nutrient availability, where Nc is reservoir specific. Constructed from: Dou et al. (2019).

Difference in HABs in different parts of the reservoir is often a result of structural variation, with nutrients and light co-limiting to various degrees in different locations. This effects not only rate of production, but also how phytoplankton adapt to their environment. Areas of the reservoir with higher light availability will be more prone to higher counts of cyanobacteria than well mixed, turbid areas (Mitrovic et al., 2011; Dantes et al., 2011). Riverine like systems are prone to a general paradigm for light limitation, as shown in Figure 2 (Dubourg et al., 2015; Tse et al., 2015, Hart, 1990). Up-river portions of the reservoir tend to be turbid and well mixed, with low light availability and low chlorophyll content (Dubourg et al., 2015; Hart, 1990). In the transitional zone (mid-reservoir), primarily production will typically be the highest (Dubourg et al., 2015; Tse et al., 2015; Hart, 1990). Management strategies should aim to target this area. Down-river portions of the reservoir are characterized by greater water depth, slower flow rate, greater light penetration, and constant sedimentation (Tse et al., 2015). Although a higher light environment is present, chlorophyll levels are typically lower in concentration than those found in the transitional zone (Dubourg et al., 2015; Hart, 1990). Here, nutrients become more limiting. Additionally, in areas that are high light and low nutrients, ratios of particulate organic carbon to Chlorophyll-a are often altered (Dubourg et al., 2015; Tse et al., 2015).

Light availability is the driving factor for growth during non-peak bloom months (Han et al., 2021) or in periods of deepest mixing (Guildford, 2000). During periods of persistent stratification, light typically exceeds amount required for growth and nutrient availability become more important (Han et al., 2021). During this time, phosphorous often becomes the limiting factor (Dubourg et al., 2015; Guildford et al., 2000). If persistent stratification occurs, water column processes (grazing, mineralization, redox, etc.) become more important than light availability. If phosphorus is efficiently recycled within a system, a decrease in nitrogen availability and shift in dominance to nitrogen fixing cyanobacteria can occur (Guildford et al., 2000). More information regarding light limitation and its effect on phytoplankton can be found in Section 3.

2.3 Balancing hydrodynamic operations with nutrient control

When altering hydrodynamic conditions within a reservoir, their effects on nutrient dynamics must be considered. If used correctly, hydrodynamic adaptive management strategies can decrease nutrient levels, subsequently reducing extent of HABs.

When determining the opportunities to influence nutrient dynamics, the placement of the outlet must be taken into consideration. Nutrient concentrations are typically best controlled with shallow (5-15 meters from bottom) water elevations (Carr et al., 2020) while hypolimnetic withdrawals are ideal to flush out nutrients released from sediments during peak stratification periods. The use of hypolimnetic withdrawals can reduce total phosphorus, total nitrogen, and total iron concentrations significantly (Dunalska et al., 2007). Over years, this method can decrease legacy nutrients in sediments and improve overall ecosystem health. Hypolimnetic

withdrawals are ideal for lakes with a high enough water budget to release during summer and fall months. This is a very low cost, low maintenance method of nutrient control. By reducing internal phosphorus loading, hypolimnetic withdrawals treats the cause of much of the water-quality deterioration in eutrophic stratified lakes (Nurnberg, 2020). However, when determining the height at which to withdraw water the effect on temperature, dissolved oxygen levels, and outflow properties must also be considered. Hypolimnetic withdrawals come with the tradeoff of increased phosphorus, nitrogen, and cold-water pollution downstream, as well as increased water temperature within the reservoir (Weber et al., 2017; Nurnberg, 2007). In order to balance dissolved oxygen levels, temperature, nutrient concentrations, and reservoir operations, mid-level withdrawals (15-25 meters) should also be considered (Carr et al., 2020) if the local infrastructure allows.

Balancing reservoir operations with release of legacy nutrients from sediments (mainly phosphorus) is also an important trade off that must be deliberated (Xu et al., 2021; Nurnberg, 2007). Releasing water at a higher velocity can potentially cause more disturbances and promote excess phosphorus release from sediments, while lower velocity releases facilitate phosphorus retention. Additionally, variations in water level elevation can effect sediment nutrient release (Xu et al., 2021; Jeppesen et al., 2015). However, controlling water level and flow rate have phosphorus economic costs that need to be taken into account. A tradeoff needs to be considered between water supply and sediment nutrient release (Xu et al., 2021).

In many cases, additional methods should be coupled with hydrodynamic control to further reduce nutrient levels. These include aeration, controlled floating wetlands, chemical inactivation of P in bottom sediments, and dredging of bottom sediments (Dunalska et al., 2007). In certain systems, particularly shallow lakes where temperature stability is hard to maintain, withdrawals may not be as effective. Instead, chemical addition, such as an aluminum compound that can adsorb and precipitate phosphorus, may be more effective (Nurnberg, 2007).

2.4 Controlled floating wetlands

Laboratory and smaller scale mesocosm experiments typically show success of controlled floating wetlands (CFWs) at removal of total nitrogen and total phosphorus (Jones et al., 2017; West et al., 2017). However, these test environments are more controlled, have fewer natural fluctuations, and typically are only carried out for a short period (Bi et al., 2019). When the system is first introduced, nutrient uptake is the highest, and decreases as plants reach maturity. This could potentially lead to overestimating of effectiveness if experiment is not carried out for a long enough period of time (Chua et al., 2012). Controlled experiments have showed the potential for CFWs to reduce Chlorophyll-a levels significantly, mainly through the uptake total nitrogen and total phosphorus. However, this also can lead to increased DOC levels in the water (Jones et al., 2017; West et al., 2017).

Removal efficiencies of CFWs are heavily influenced by temperature (Bi et al., 2019; Wang and David, 2014; Van de Moortel, 2010). Species selected for CFWs typically have a higher removal efficiency during growing season (autumn/summer), and lower biological activity during the colder months. This is due to the fact that macrophytes grow faster at higher temperatures, leading to increased pollutant uptake. This could increase their importance as climate changes and water temperatures increase (Bi et al., 2019). CFWs will need removal at the end of the growing season to avoid re-release of large amount of nutrients into the water, and the system will need replaced again the next year (Jones et al., 2017). An appropriate growth rate of plants must be chosen for the specific system. If growth is too high, frequent harvesting and maintenance will be required, which is not practical for large scale systems. A plant with moderate growth rate, high nutrient uptake, with the ability to survive in the chosen environment must be selected (Chua et al., 2012).

While there have been several small-scale studies on CFWs, less is known about their efficiency in a real world situation. Removal efficiency of CFWs varied significantly between systems, and more in-situ studies are needed to determine the success of this nutrient removal method (Bi et al., 2019). Calculation of size of CFWs needed to make a significant impact: water body size can be excessive, which is not practical for many systems. The use of coupling CFWs with other management strategies could allow for less CFW area coverage, but more studies are needed to be done on this (Chua et al., 2012). Floating design of CFWs leaves the potential for this nutrient management method to be combined with hydrodynamic management, such as pulsed inflows or withdrawals (Bi et al., 2019). More research needs to be done in order to determine how these smaller scale experiments can be scaled up to real world situations, particularly how wetland: water volume would need to be modified in order to efficiently reduce phytoplankton blooms in larger lakes (Jones et al., 2017). Although they have long-term costs, CFWs do have the potential to successfully reduce nutrient levels, especially in conjuncture with other management efforts.

2.5 Summary of nutrient control

While most recent management strategies have been focused on reduction of total phosphorus, reduction of total nitrogen is also important in many systems, particularly those with high legacy phosphorus. Which nutrient is limiting and to what extent is specific to the type of lake, extent of eutrophication, redox reactions occurring, and type of HAB. Varying nitrogen to phosphorus ratios has the potential to alter dominant species of phytoplankton. Light can also affect nutrient dynamics, especially in deeper, well mixed portions of the reservoir, and during periods of destratification.

Altering hydrodynamic conditions, based on the specific characteristics of a system, can lead to reduction of water column and/or legacy nutrients. These management methods should be carried out with attention to consequences other than nutrients control. For example,

hypolimnetic withdrawals are an effective, low-cost method to reduce bottom nutrients released from sediments following persistent stratification. However, this often comes with the trade-off of decreased water quality of outflow, increased temperature within the reservoir, and less water available for hydropower generation. Each of these things must be considered when determining which nutrient management strategy is correct for a system. Controlled floating wetlands have been shown on smaller scales to successfully reduce nutrient levels and control blooms. However, more research is necessary to determine how these experiments can be scaled up to real world situations.

3. Seasonal Limnology

3.1 Stratification

Stratification is influenced by seasonal weather patterns, with most persistent stratification occurring in warm, calm conditions, when flow rate is low (Xu, 2021). The extent of stratification is diurnal, with more intense conditions occurring during the day, and lessening at night (Bormans et al., 1997). Higher stratification correlates with higher cyanobacterial concentration (Dantas et al., 2011; Mitrovic et al., 2003; Bormans et al. 1997). More prolonged and earlier onset stratification is expected in the coming decades (Sahoo et al., 2016). With this, reduced mixing, lower dissolved oxygen and increased sediment nutrient release are likely to occur. Understanding how stratification alters bloom dynamics is key to understanding how to combat HABs.

Relationships between stratification and likelihood of bloom occurrence have been established both through analysis of historical data and model simulations. Extent of stratification is typically determined using a combination of observational data, flow rate, daily maximum air temperature, and the difference between maximum and minimum daily water temperature (Viney et al., 2007; Mitrovic et al., 2003). Models that have been modified to establish relationships between stratification and bloom occurrence include the CE-QUAL-W2 model (Xu et al., 2021), the Princeton Ocean Model (POM) (Bormans and Condie, 1997), and the Integrated Quantity-Quality Model (IQQM) (Mitrovic et al., 2011; Viney et al., 2007). For both direct sampling and model data, plotting of stratification in comparison to phytoplankton concentrations has been found to result in the general trend that blooms are favored during periods of persistent stratification.

During peak stratification periods, cyanobacteria, such as *Anabaena*, *Cylindrospermopsis* or *Microcystis*, are favored over diatoms and dinoflagellates (Dantas et al., 2011; Jungo et al., 2001; Reynolds, 1976). Main contributors to cyanobacterial dominance are optimal light irradiance, optimal temperature for growth, and nutrient storage capability. Additionally, stratification often favors the growth of species with buoyancy regulation (Winder and Hunter,

2008; Reynolds, 1976). Cyanobacteria are positively buoyant, allowing them to float near the surface during non-mixing periods. During these periods, there is a high loss of negatively buoyant diatoms via sedimentation due to sinking rate (Visser et al., 2016; Bormans and Condie, 1997). When mixed, negatively buoyant species can remain in the euphotic zone for longer and compete for nutrients (Roelke and Pierce, 2011; Bormans and Condie, 1997). Nutrients levels present in inflow can also drastically effect community composition. For instance, inflow rich in nitrate has shown a shift to favor diatoms (Roelke and Pierce, 2011). During periods of persistent stratification and low flow rate, phytoplankton with the ability to store nutrients, such as *Cylindrospermopsis raciborskii*, dominate (Dantas et al., 2011). Additionally, different species of phytoplankton are more sensitive to light. High light levels lead to higher cell counts of cyanobacteria (Xiao et al., 2017; Mitrovic et al., 2011). During periods of destratification, cyanobacteria are pushed below the surface, decreasing their light availability while negatively buoyant species are brought to the surface and increase competition, which can lead to a shift in community composition (Heo and Kim, 2004; Hawkins and Griffiths, 1993). Other factors that contribute to cyanobacterial dominance include growth rate, oxygen availability, pH, and temperature. Overall, persistent stratification reduces competition, decreases diversity, and leads to cyanobacterial dominated blooms. Destratification leads to a more diverse phytoplankton community structure, typically composed of diatoms and phytoflagellates in addition to cyanobacteria, preventing singular cyanobacterial dominance.

Many argue that breaking up stratification during peak periods is an effective method to suppress HABs (Li et al., 2021; Xu et al., 2021; Mitrovic et al., 2011; Viney et al., 2007). This can be done through altering flow rate, pulsed inflows, or artificial mixing, as discussed in Section 1. These methods are typically done during the warm and/or drought season, when peak blooms occur with the objectives of inducing a significant change in hydrodynamic conditions, altering nutrient dynamics, and light availability to decrease the competitive edge of cyanobacteria and ultimately reduce blooms. Increasing inflow can effectively break up stratification and increase turbidity within the water column (Roelke and Pierce, 2011). Increasing discharge during periods of persistent stratification has been found to be an effective method of controlling algal blooms (Li et al. 2012; Mitrovic et al., 2011). However, flow cannot be maintained indefinitely, and it must be determined what level and duration of flow is needed to de-stratify the system and prevent blooms from occurring. Mixing can increase oxygen levels, increase temperature of deeper layers, decrease surface temperature, alter nutrient availability, change pH, and shift composition from cyanobacteria to green algae/diatoms (Visser et al., 2016). Additionally, mixing entrains buoyant cyanobacteria where less light is available, decreasing their growth (Ma et al., 2015; Lian et al., 2014). When mixed, light becomes limiting, and algae better adapted to fluctuating light levels begin to co-dominate. However, mixing is not always effective against blooms. In order to be successful, mixing rate must be high enough to entrain cyanobacteria and decrease their light exposure. Mixing was found most effective at preventing blooms in deeper systems (> 15 meters) (Visser et al., 2016).

While many believe that breaking up stratification is key to suppressing HABs, there is also significant evidence that destratification can actually worsen bloom conditions. Breaking up stratification does not work for shallower systems (< 15 meters), since water is not deep enough to cause light limitation (Visser et al., 2016). If hypolimnetic nutrient levels are high, destratification can lead to higher nutrient availability at the surface, adversely affecting surface water quality (Bormans et al., 2015; Nurnberg, 2007). While mixing may result in a shift from cyanobacterial dominance to diatoms or flagellates, it may not decrease total algal concentration overall (Heo and Kim, 2004). In some cases, this influx of nutrient availability to the surface can actually increase algal biomass (Jungo et al., 2001; Hawkins and Griffiths, 1993). Breaking up stratification also can lead to significant temperature increase in the hypolimnetic layer, which can have adverse effects on water quality and aquatic life (Nurnberg, 2007). Before considering breaking up stratification as an adaptive management strategy, these factors must be taken into account. If maintaining thermal stability is desired, surface flushing or less vigorous hypolimnetic withdrawals should be deliberated.

3.2 Temperature

In order to accurately capture the effects of seasonal influence on phytoplankton dynamics, temperature is a critical component to consider, especially when looking at modelling phytoplankton growth (Canale et al., 1974; Grimaud et al., 2015; Fanesi et al., 2017). There is a strong dependence of phytoplankton growth rate on temperature (Nalewajko and Murphy, 2001; Fanesi, 2017). While higher water temperatures support higher and faster growth, other factors must also be considered for bloom development. The most vulnerable areas to HAB events are those that experience higher temperatures and lower flow rates (Xin et al., 2020), as well as have readily available nutrients (primarily phosphorus) (Nalewajko and Murphy, 2001). Cyanobacteria have the ability to adapt to increased water temperatures and strong light levels. High intraspecific variation indicates that multiple strains of cyanobacteria drive adaptation to various environmental conditions, especially temperature and light levels (Xiao et al., 2017). The stronger the temperature stratification, the more likely the occurrence of cyanobacterial blooms (Chuo et al., 2019). Both *Microcystis* and *Anabaena* have been associated with hot, dry weather and low N:P ratios (Wang et al., 2016; Nalewajko and Murphy, 2001). Two ways that water resource managers can combat temperature influence are varying flow or using a temperature control curtain.

Since the most vulnerable areas to blooms are those which experienced lower flow rate and higher temperatures, increasing flow rate during periods where temperature stratification is strongest will have the maximum effect of reducing HABs (Chuo et al., 2019). Flow rate typically can be varied through the operation of cascade dams, drawdown, or releases from secondary reservoirs. Looking at forecasted daily air temperature for days where temperature is expected to be continuously higher than the 5-day mean value of the corresponding period, and increasing flow rate, has shown past success (Xin et al., 2020). Another way to target increased

flow rate is to look at difference between minimum and maximum water temperature (Viney et al., 2007). When minimum and maximum temperatures are close in value, mixing is likely to occur, and blooms are less likely to develop. When minimum and maximum temperature in the water column has a greater difference, mixing will be weaker, stratification will most likely occur, and conditions will be favorable to blooms. This is when flow rate should be increased. Varying flow rate is a good low-cost, short-term strategy for combatting HABs within a system.

Thermal control curtains (TCC) are primarily used to facilitate selective withdrawals within a reservoir to change the physical and chemical properties of the outflow. TCC scenarios have shown the ability to decrease Chl-a levels (He et al., 2019). TCCs are designed using an impermeable fabric to block flow at a certain depth, which forces water to flow either above or below the curtain (Lian et al., 2019). There are two major types of TCC, which produce significantly different outflow and interior reservoir properties. The top-TCC, which places a curtain at the top and facilitates high water flow below the epilimnion; and the bottom-TCC, which places a curtain at the bottom, restraining vertical flow between the epilimnion and hypolimnion, causing the reservoir to behave similar to a river channel (He et al., 2017). Understanding why and when to use top versus bottom TCC is crucial. Typically, top-TCC is recommended in the summer/fall and bottom-TCC in the winter/ spring (Bartholow et al., 2001). The use of the top-TCC deepens the thermocline, reduces stability of the water column, reduces seasonal variations in temperature outflow, and creates a high temperature difference between inflow and outflow. This is practical for water quality improvement, decreasing stability in the reservoir, and generating cooler summer outflow temperatures. The use of bottom-TCC makes the thermocline shallower, amplifies seasonal variation in outflow temperature, and reduces temperature difference between inflow and outflow. This is practical to restore natural inflow temperatures downstream (He et al., 2017). Bottom-TCCs have an annual overall effect of decreasing Chl-a levels within a reservoir (He et al., 2019). A combination of both TCC scenarios should be varied seasonally based on specifics of the system in order to maximize management efficiency. It should be noted that temperature control devices are more effective at controlling properties of the outflow, rather than characteristics of the reservoir itself. While the TCC had a significant effect on stratification, it had no dramatic effect on nutrient composition (Bartholow et al., 2001). TCCs should be coupled with other management strategies to maximize effectiveness. More information regarding the construction and visualization of the two types of temperature control curtains can be found in He et al., 2019 and Lian et al., 2019.

A few other methods can be considered when looking to control temperature. The depth outflow is taken from can be crucial for determining temperature profile within a system. When water is taken from the bottom, the thermocline deepens and water column stability decreases, breaking up stratification (Carr et al., 2020). Increasing water level strengthens vertical movement, which reduces algal time in the euphotic zone, which decreases the photosynthetic rate, thus controlling blooms. A sufficiently large water level fluctuation can disrupt seasonal stratification, influence vertical light and heat conditions, and control algal

blooms. Analyzing the influence of water level fluctuations on algal blooms is key for the effective implementation of the reservoir operations (Chuo et al., 2019). Aquatic vegetation, in addition to reducing nutrient and light availability, can be used to reduce water temperature. Since trees typically require more time to grow to size large enough to shade and reduce temperature, this is considered a long-term management strategy. (de Winton et al., 2013; Hutchins et al., 2010). This is a cost-effective method that could potentially be coupled with hydrodynamic controls, depending on the structure of the reservoir. Since many toxic strains of algae thrive in high temperatures, it is important to continue exploring and developing methods of temperature control.

Climate simulations have indicated that the occurrence and duration of blooms, especially those who thrive in high temperature environments, are expected to increase in years to come, as climate becomes hotter. Increasing temperature and decreasing amount of water available to discharge leads to an increase in frequency and duration of HABs (Viney et al., 2007). For stratified reservoirs, increased cyanobacterial cell counts coincide with earlier and longer summer warming of surface waters and greater summer precipitation (Smucker et al., 2021). More intense and prolonged stratification favors cyanobacteria over other phytoplankton species because cyanobacteria have the ability to regulate their buoyancy to acquire nutrients and light in the surface layer. A study in Japan found abundant concentrations of *Anabaena* and *Microcystis* when maximum temperatures reached 32 degrees Celsius. When maximum temperatures reached only 28 degrees Celsius, these two species did not appear (Nalewajko and Murphy, 2001). Yearly trends in temperature are important to consider in reservoir management as climate continues to change.

3.3 Light

Cyanobacteria species prone to blooms can survive in both low light and high light environments. This is due to the presence of multiple strains of each species, with varying abilities to respond to different environmental conditions (Islam and Beardall, 2017; Xiao et al., 2017). Often, multiple strains of cyanobacteria are present, driving adaptation to various environmental conditions. For example, *Microcystis aeruginosa* were found to have higher growth rates at high light intensity and low temperature, straight *Cylindrospermopsis raciborskii* showed highest growth rates at low light intensity and high temperature, and coiled *Cylindrospermopsis raciborskii* showed highest growth at high light intensity and high temperature. (Xiao et al., 2017). Both *Microcystis* and *Anabaena* survive in various light climates, with toxic and non-toxic strains changing dominance based on light intensity. Toxic strains had a higher growth rate at higher light intensity than non-toxic species and tend to dominate when light is readily available (Islam and Beardall, 2017). Overall mean growth rate of cyanobacteria increases with increasing light and temperature (Xiao et al., 2017). One reason that cyanobacteria, such as *Microcystis*, dominate in high light environments is because their buoyancy regulation allows them to move through the water column and find light conditions

most favorable to growth (Chorus and Bartram, 1999). A strong correlation can be found between weather patterns, such as solar radiation and cloud cover, and algal growth (Cao et al., 2011). Both nutrient limitation and shading are important factors to reduce algal blooms. However, cutting nutrient loads from agricultural input is often not feasible in a real-world situation, and typically out of control of those managing the reservoir systems. Light limitation is a more cost effective, attainable way to reduce algal blooms (Hutchins et al., 2010).

Similar to nutrients, light conditions can be strongly influenced by the alteration of hydrodynamics. Hydrodynamic conditions were found to be the dominating factor in algal growth and community composition (Lian et al., 2014; Rigosi and Ruenda, 2012). Systems with high light levels and low turbidity coincide with high algal cell counts (Mitrovic et al., 2011). When mixed, cyanobacteria are pushed into deeper water, where they no longer receive light and die (Lian et al., 2014; Mitrovic et al., 2011). Additionally, mixing allows for negatively buoyant species to vertically migrate and compete for nutrients in the photic zone (Visser et al., 2016; Mitrovic et al., 2011). Potential methods of hydrodynamic control are discussed in more detail in Section 1. In addition to altering hydrodynamics, direct shading and photocatalytic inactivation can also be used to control blooms. These are discussed in more detail in the following paragraphs.

TiO₂ coated balls have been successful at reducing concentrations of Chlorophyll-a and phycocyanin, through direct light limitation and photocatalytic inactivation (Lee et al., 2020; Kim et al., 2018; Kim and Lee, 2005). Algae float vertically to the surface when light levels decrease, leading them to contact the TiO₂ balls. These balls kill cyanobacteria through the use of highly reactive hydroxyl radicals, which degrade the lipid cell membrane of cyanobacteria through peroxidation reduction (Kim et al., 2018). This causes them to lose their photosynthetic ability (Kim and Lee, 2005). This method reduces cyanobacterial blooms more effectively than diatom blooms, since the silica cell wall on diatoms prevents them from being effective to such a large extent (Lee et al., 2020; Kim and Lee, 2005). This method is more effective at reducing blooms in shallow systems than in deeper ones (Lee et al., 2020). Some pros to this method are cost effectiveness, practicality, and high effectiveness. Some cons include that their effectiveness decreases over time, they must be replaced yearly, limit recreational activities, and if made improperly they could leach nanoparticles into the water. Additionally, the decaying algae remains in the water body, which re-releases nutrients back to the water column.

Enclosures, such as square sections or pontoon type, can be used to block sunlight, effectively getting rid of weather dependence. This method works through direct shading, decreasing light available to HABs, and subsequently decreasing their growth rate. Enclosures have been successful in reducing algal cell densities and chlorophyll-a concentrations (Cao et al., 2011; Li et al., 2011). This method is only effective if the shade is in place during peak periods, which can be difficult for larger systems. Rapid algal growth rate was observed when shade was removed too early (Li et al., 2011). Additionally, since decreased plant and algae growth means less

nutrient uptake, this method often leads to high levels of nutrient accumulation in the water (Cao et al., 2011; Li et al., 2011).

Shading from vegetation has proven to be an effective way to reduce temperature, light availability, and algal blooms (Burrell et al., 2014; Yeh et al., 2011; Hutchins et al., 2010). This method works both through reducing solar penetration and increasing competition for nutrients. When looking at multiple systems with varying nutrient inputs and level of riparian cover, it was found that gross primary production decreased with increased riparian cover. Shading was found to have a stronger correlation with primary production than agricultural input, land use, or local characteristics (Burrell et al., 2014). Riparian shading is often a good alternative for physical enclosures, which are not feasible for larger scale operations (Yeh et al., 2011). However, type and location of vegetation must be determined for each specific system to assess effectiveness. Other cons to this method include interference with recreation, debris from trees could pose a safety issue, and that this is a long-term operation (de Winton et al., 2013).

Floating solar photovoltaics are a fast-growing technology that have emerged as an electricity option that bypasses many land use issues (Gonzalez Sanchez et al., 2021). These systems may have the additional advantage of improving water quality through reducing algal growth. Use of floating photovoltaic plants has shown potential to decrease algal bloom concentrations via light limitation (Haas et al., 2020). The amount of reservoir cover must be determined to balance reduction of chlorophyll-a levels, without reducing revenue too significantly. Some pros to using these systems is it lowers overall cost of using solar panels (eliminates cost of renting land), reduces evaporation of the water body, limits light availability to reduce blooms, naturally cools down the panels, increasing their efficiency, and reduces CO₂ emissions. Some cons include restricting amount of reservoir available for recreational activities, lowers concentration of non-HAB/ important primary producer species, and the panels must be built to withstand storms. Floating solar photovoltaics are an emerging technology, with further research needed to address the effects on environmental factors.

3.4 Water level fluctuation

Water level fluctuations alter the structure of the reservoir, light availability, and nutrient dynamics. In riverine systems, water level fluctuations were found to be more important in predicting algal cell biomass than nutrient concentration or flow velocity (Xia et al., 2020). Water level is often altered seasonally through impoundment processes during flood season and drawn down during periods of drought. Additionally, most adaptive management strategies used to control blooms also have significant impact on water level. Thus, when employing the hydrodynamic control methods discussed in Section 1, the effects on water level within the reservoir must also be considered. Analyzing the influence of water level fluctuations on algal blooms is key for the effective implementation of the reservoir operations (Chuo et al., 2019).

Water level can be decreased for a variety of reasons, such as flushing sediment or altering flow rate downstream. Significant reduction in water level can lead to conditions that are often more favorable to larger and more frequent cyanobacterial blooms (Jeppesen et al., 2015). Blooms were found more likely to occur when water level was significantly lower than the previous ten days (Xia et al., 2020). When water level is decreased, there is increased exposure of the littoral zone to sediments, which can lead to increased phosphorous release, and increased vulnerability to species that interact with the sediment layer (Shin et al., 2022). If wind induced mixing is strong enough, this can lead to increase nutrient availability to the surface, further worsening bloom conditions. Significant depth decrease can cause clear, macrophyte-dominated lakes to become turbid and phytoplankton dominated, often by toxin producing species (Jeppesen et al., 2015; Zohary and Ostrovsky, 2011). With the warming associated with decreased water level, species such as *Microcystis spp.*, *Cylindrospermopsis raciborskii*, *Chrysosporum ovalisporum*, or *Dolichospermum spp.* are more likely to thrive (Xiao et al., 2017).

Water level in reservoirs can also be increased for various purposes, such as flood control or storage. An increase in water level decreases light penetration to the bottom of the system, increasing the vulnerability of non-surface species (Jeppesen et al., 2015). Higher water level has been found to be helpful in reducing bloom extent (Lian et al., 2014). However, excess rise in water level in some cases can lead to increase in invasive species, over-population of fish, and zooplankton decline (Zohary and Ostrovsky, 2011).

Bloom dependence on water level fluctuations is more important in shallower systems, due to their high surface area to volume ratio. (Xia et al., 2020; Jeppesen et al., 2015). Excess water level fluctuations, as well as severely reduced fluctuations, are known to impair ecosystem functioning in shallow systems (less than fifteen meters deep). This includes increase in nuisance species, loss of biodiversity, increased nutrient loading, and larger and more frequent cyanobacterial blooms (Zohary and Ostrovsky, 2011). Intermediate fluctuations are most suited to maintain ecosystem functioning, generate environmental change, and reduce blooms dominated by a specific species (Rigosi and Ruenda, 2012; Zohary and Ostrovsky, 2011; Padisak et al., 1999).

Temperature and precipitation patterns are expected to change in the coming decades (Smucker et al., 2021; Sahoo et al., 2016). As meteorological patterns change and water demand increases, effects of varying water level must be considered. Higher demand for agricultural and drinking water are expected as population increases. Prolonged residence time of water as a result of drought conditions typically favors cyanobacterial blooms, especially *Microcystis* (Jeppesen et al., 2015). Larger *Microcystis* populations are also associated with hot, dry weather and areas of low water level (Nalewajko and Murphy, 2001).

3.5 Wind

Wind is often the driving force of downward heat transport, and influences intensity and duration of stratification. During weak wind events, reservoirs are more likely to experience sharper temperature gradients, increased thermal stability, and delayed destratification (Zhang, et al., 2020). In recent decades, terrestrial wind speeds have been decreasing, leaving many lakes more susceptible to worsening bloom conditions (Huang et al., 2020; Zhang et al., 2021).

Wind driven hydrodynamic conditions are often an important factor in extent of cyanobacterial blooms (Wu et al., 2015). Cyanobacteria are typically favored under warm, calm conditions, while other species dominate in cold or windy conditions. There has shown to be significant negative correlation between wind speed and floating algal bloom area (Zhang et al., 2021), as well as surface *Microcystis* concentrations (Cao et al., 2011). In warmer months, the negative relationship between blooms and wind speed becomes stronger (Zhang et al., 2021). In deeper lakes, decreasing wind speed can lead to early onset and/or prolonged periods of stratification. This, in turn, can lead to increased release of legacy nutrients from sediments. In shallow lakes, decreased wind speed also favors increased stratification, but less is known about the effect on nutrient dynamics (Zhang et al., 2021).

Turbidity often increases following observations of high wind gusts (Elci, 2008). Turbidity can be used to improve descriptions of light environment, subsequently producing a more accurate analysis of algal production (Brown, 1984). Higher turbidity reduces light availability and limits algal growth (Dubourg et al., 2015; Mitrovic et al., 2011). The dominant wind direction, bathymetry, and inflow conditions can be combined to determine which locations within a lake are most susceptible to algal blooms (Zhang et al., 2021). Changes in daily surface flow direction can be driven by periodic wind speed (Xu et al., 2021). Typically, 3 m/s wind speed is sufficient to prevent formation of surface blooms (Cao et al., 2011; Zhang et al., 2021). This value can be used as an initial warning threshold for peak bloom periods but should be adapted for each specific lake. Since shallower water depth is more favorable for cyanobacteria to float to the surface and form blooms, shallower systems are more sensitive to wind changes than deeper systems (Zhang et al., 2021).

Conclusion

Harmful algal blooms are an ongoing challenge to the management of reservoir water quality that will only continue to worsen in the coming years. Those in charge of reservoir water resources must be proactive and proficient in keeping blooms under control. Determining what method is best suited for each individual system can be complex but is paramount to optimizing water quality.

Changing hydrodynamic conditions can be an attainable method of controlling algal blooms on a short timescale. These methods work through changing conditions within the reservoir itself.

Surface withdrawals are ideal for reducing surface HAB biomass and surface nutrients, maintaining colder hypolimnion temperature, and reducing temperature difference between inflow and outflow. Bottom withdrawals are ideal for reducing legacy nutrients, removing oxygen deficient water, and providing cooler outflow. Pulsed inflows are ideal for riverine systems, where water from upstream can be used to dilute blooms downstream during peak periods. Artificial mixing has been successful in some deeper systems at reducing overall phytoplankton biomass, as well as shifting dominance from cyanobacteria to less harmful species. However, artificial mixing shows high variability between systems. Each of these methods should be used with caution, especially in regard to proper timing, fluctuations in water level, and impact on overall ecosystem health.

Management of nutrient input is another key factor of controlling HABs, especially as population continues to grow, increasing demand for agricultural practices and subsequent nutrient input. While limiting input of nutrients would be ideal, this is often out of the reservoir manager's scope but can be achieved with integrated watershed management. Determining whether phosphorus or nitrogen is limiting can greatly aid in water quality management. Which nutrient is limiting, and to what extent, is specific to the type of lake, redox reactions occurring, and type of phytoplankton. Co-limitation between nutrients and light availability, which varies seasonally and with structure of the reservoir, is also important to consider. Changing certain hydrodynamics within the reservoir can help decrease nutrients already present. Hypolimnetic withdrawals works by decreasing legacy phosphorus released from sediments under anoxic conditions and have shown success at decreasing overall surface phosphorus levels. Controlled floating wetlands have shown on smaller scales to successfully reduce nutrient levels, however, further research needs to be done to determine if these experiments can be scaled up to real world situations.

How the reservoir environment changes with seasonal variation is also crucial to keeping HABs under control. Stratification is influenced by seasonal weather patterns, with most persistent stratification occurring in warm, calm conditions, when flow rate and wind is low. Breaking up stratification can induce significant hydrodynamic changes, altering nutrient dynamics, light availability, and community composition. This can be done through altering flow rate, pulsed inflows, or artificial mixing. However, breaking up stratification can sometimes worsen water quality. In cases where it is ideal to maintain thermal stability, surface flushing or less vigorous hypolimnetic withdrawals should be used. Determining whether stratification should be retained or broken up is critical to successful management of blooms. Toxic strains of cyanobacteria have a higher growth rate at high light levels, so light limitation can be used to decrease blooms. This can be done through changing hydrodynamic conditions, the use of TiO_2 coated balls, enclosures, riparian shading, or floating photovoltaic plants. Cyanobacteria also thrive in high temperature environments. When temperature is high, flow rate should be increased to create conditions less favorable to blooms. Thermal control curtains can be used to change withdrawal height, altering physical and chemical properties of outflow, as well as

within the reservoir itself. Bottom versus top withdrawals should be varied seasonally for optimal results.

In the coming decades, temperature, precipitation, nutrient levels, and flow rate patterns are projected to change. Understanding the impact of these variations on quantity and quality of water resources is crucial. There are countless distinct freshwater bodies around the globe, and each will respond differently as climate changes. To evaluate the relative impacts of different management strategies on each specific system, the use of water quality modeling is advantageous and should thus be explored as a management investment. In the face of increasing drivers for bloom conditions, proficient and proactive management of reservoirs can significantly reduce bloom occurrence and improve overall ecosystem health.

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